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More on the

functor of points

Last time The tensor product of spectra, review of the functor of points approach to algebraic geometry

Today More on the functor of points, quotients

> Recall from last time

Thm The ∞ -category Sp admits a unique symmetric monoidal structure $\otimes = \otimes_S$ satisfying

(1) S is the unit

(2) $\otimes$ preserves colimits separately in each variable

Moreover, this symmetric monoidal structure also satisfies,

(3) The functor $\Sigma^\infty: (\mathrm{An}_*, 1) \longrightarrow (\mathrm{Sp}, \otimes)$ is symmetric monoidal.

(4) This symmetric monoidal structure is closed.

Thm The functor

$$\begin{array}{ccc} \text{Sch} & \longrightarrow & \text{Fun}_{\text{zar}}(\text{CAlg}, \text{Set}) \\ X & \longmapsto & X(-) \end{array}$$

is fully faithful with image those Zariski sheaves F such that there exists a collection of open subfunctors $(U_i)_{i \in I}$ such that.

(1) Each U_i is representable, i.e., $U_i \simeq \text{Map}_{\text{CAlg}}(B_i, -)$ for some ring B_i .

(2) The induced map $\bigsqcup_{i \in I} U_i \longrightarrow F$ is an epimorphism in $\text{Fun}_{\text{zar}}(\text{CAlg}, \text{Set})$.

Faithfully flat descent

Thm (fpqc descent) Let X be a scheme. Then for every faithfully flat quascompact map $T \rightarrow S$, the natural diagram

$$\text{fidèlement plat} \quad X(S) \longrightarrow X(T) \rightrightarrows X(T \times_S T)$$

in French

is an equalizer

Def. A functor $F: \text{Sch}^{\text{op}} \rightarrow \text{Set}$ is an fpqc sheaf if F is a Zariski sheaf and for every faithfully flat quascompact map $T \rightarrow S$, the natural diagram

$$F(S) \longrightarrow F(T) \rightrightarrows F(T \times_S T)$$

is an equalizer

> Restricting to CAlg , Zariski descent is automatic.

A functor $F: \text{CAlg} \rightarrow \text{Set}$ is an fpqc sheaf if for every faithfully flat ring hom $A \rightarrow B$, the natural diagram

$$F(A) \longrightarrow F(B) \rightrightarrows F(B \otimes_A B)$$

is an equalizer.

Cor. Restriction along $\text{Spec} : \mathbf{CAlg} \hookrightarrow \mathbf{Sch}^{\text{op}}$ defines an equiv. of categories

$$\mathbf{Sh}_{\text{fpqc}}(\mathbf{Sch}, \mathbf{Set}) \xrightarrow{\sim} \mathbf{Fun}_{\text{fpqc}}(\mathbf{CAlg}, \mathbf{Set})$$

Quasicoherent sheaves via the functor of points

Note. The assignment $A \mapsto \text{Mod}_A^{\heartsuit}$ and

$$[\phi: A \rightarrow B] \mapsto [B \otimes_A (-): \text{Mod}_A^{\heartsuit} \rightarrow \text{Mod}_B^{\heartsuit}]$$

can be refined to a functor

$$\text{CAlg} \longrightarrow \text{CMon}(\text{Cat}_1)$$

to symmetric monoidal 1-categories

Observe. If $\mathcal{C}$ is an ∞ -category with finite products, then since limits commute, $\text{CMon}(\mathcal{C}) \hookrightarrow \text{Fun}(\Delta^{\text{op}}, \mathcal{C})$ is closed under all limits that $\mathcal{C}$ admits. Hence the forgetful functor

$$\begin{array}{ccc} \text{CMon}(\mathcal{C}) & \longrightarrow & \mathcal{C} \\ X & \longmapsto & X_1 \end{array}$$

preserves all limits that $\mathcal{C}$ admits

Ex $\mathbf{CMon}(\mathbf{Cat}_1)$ has limits and the forgetful functor preserves them.

Ntn we'll just write $\mathbf{Spec} \mathbf{CAlg}^{\text{op}} \hookrightarrow \mathbf{Fun}(\mathbf{CAlg}, \mathbf{Set})$ for the contravariant Yoneda embedding

Def we define

$$\mathbf{QCoh}(-)^{\heartsuit} : \mathbf{Fun}(\mathbf{CAlg}, \mathbf{Set})^{\text{op}} \longrightarrow \mathbf{CMon}(\mathbf{Cat}_1)$$

to be the unique functor that preserves limits and whose restriction along

$$\mathbf{Spec} \mathbf{CAlg} \hookrightarrow \mathbf{Fun}(\mathbf{CAlg}, \mathbf{Set})^{\text{op}}$$

is $\mathbf{Mod}_{(-)}^{\heartsuit} : \mathbf{CAlg} \longrightarrow \mathbf{CMon}(\mathbf{Cat}_1)$. Explicitly, we send X to the limit of 1-categories

$$\mathbf{QCoh}(X)^{\heartsuit} := \lim_{x: \mathbf{Spec}(A) \rightarrow X} \mathbf{Mod}_A^{\heartsuit}$$

> For each map $f: X \rightarrow Y$ in $\text{Fun}(\text{CAlg}, \text{Set})$, we write $f^*: \text{QCoh}(Y)^{\heartsuit} \rightarrow \text{QCoh}(X)^{\heartsuit}$ for the induced symmetric monoidal functor.

Note Even more explicitly, an object of $\text{QCoh}(X)^{\heartsuit}$ consists of

(1) For each $x: \text{Spec}(A) \rightarrow X$, an A -module $M(x)$

(2) For each com triangle

$$\begin{array}{ccc} \text{Spec}(B) & \xrightarrow{\text{Spec}(\phi)} & \text{Spec}(A) \\ & \searrow \gamma & \swarrow x \\ & X & \end{array}$$

an isomorphism of B -modules $B \otimes_A M_x \simeq M_y$

such that the isomorphisms in (2) satisfy the cocycle condition.

Remark (pushforward functoriality). Modulo some size issues which basically never appear in practice, for a map $f: X \rightarrow Y$ in $\text{Fun}(\text{CAlg}, \text{Set})$, the adjoint functor theorem implies that

$$f^*: \text{QCoh}(Y)^{\heartsuit} \longrightarrow \text{QCoh}(X)^{\heartsuit}$$

admits a right adjoint

$$f_*: \text{QCoh}(X)^{\heartsuit} \longrightarrow \text{QCoh}(Y)^{\heartsuit}$$

This is true, for example, if X and Y are accessible, i.e., small colimits of representables in $\text{Fun}(\text{CAlg}, \text{Set})$.

Remark. If we're given any functor $\text{CAlg} \rightarrow \text{Cat}_1$, e.g., modules of finite presentation, finite projective modules, etc., we can use the same type of definition to extend the functor to $\text{Fun}(\text{CAlg}, \text{Set})$. This is how one can, for example, define the category of vector bundles on a scheme.

Descent for quasicoherent sheaves

Question How do we know that this definition of quasicoherent sheaves agrees with the one given in the locally ringed spaces approach to algebraic geometry?

Thm (fpqc descent for quasicoherent sheaves, Grothendieck)

The functor $\mathrm{QCoh}(-)^{\heartsuit} : \mathrm{Sch}^{\mathrm{op}} \rightarrow \mathrm{CMon}(\mathrm{Cat}_1)$ satisfies fpqc descent.

(1) For any scheme X and open cover $(U_i)_{i \in I}$, the natural diagram of Sym. mon. Categories

$$\mathrm{QCoh}(X)^{\heartsuit} \longrightarrow \prod_{i \in I} \mathrm{QCoh}(U_i)^{\heartsuit} \rightrightarrows \prod_{i,j \in I} \mathrm{QCoh}(U_{i,j})^{\heartsuit} \rightrightarrows \prod_{i,j,k \in I} \mathrm{QCoh}(U_{i,j,k})^{\heartsuit}$$

$U_{i,j} := U_i \cap U_j$
 $U_{i,j,k} := U_i \cap U_j \cap U_k$

is a limit diagram.

(2) For every faithfully flat quasicompact map $Y \rightarrow X$, the natural diagram of Symmetric monoidal Categories

$$\mathrm{QCoh}(X)^{\heartsuit} \longrightarrow \mathrm{QCoh}(Y)^{\heartsuit} \rightrightarrows \mathrm{QCoh}(Y \times_X Y)^{\heartsuit} \rightrightarrows \mathrm{QCoh}(Y \times_X Y \times_X Y)^{\heartsuit}$$

is a limit diagram

Key point It is also easy to see that the locally ringed spaces definition of quasicoherent sheaves is an fpqc sheaf. Since the two definitions agree on affine schemes, we deduce that they agree in general.

Remark (the indexing category) Note that here, the sheaf condition involves an additional term. Indeed, the result is not true if we don't include the 'triple intersection' term. To align this with the usual sheaf condition, we make the following observation.

- > For $n \geq 0$, write $\Delta_{\leq n} \subset \Delta$ for the full subcategory spanned by objects isomorphic to $[0], \dots, [n]$.
- > Write $\Delta^{inj} \subset \Delta$ for the non-full subcategory with all objects, but only the injective maps. Write $\Delta_{\leq n}^{inj} = \Delta_{\leq n} \cap \Delta^{inj}$.

> Usually, the sheaf condition is formulated as a limit of a diagram over $\mathbb{A}_{\leq 1}$ or $\mathbb{A}_{\leq 1}^{inj}$.

Definition Let $n \geq 0$.

(1) An anima X is n -truncated if for all $i > n$ and $x \in X$, the group $\pi_i(X, x)$ is trivial.

(2) An ∞ -category $\mathcal{C}$ is an n -category or an $(n, 1)$ -category if all of its mapping anima are $(n-1)$ -truncated.

Write $An_{\leq n} \subset An$ for the full subcategory spanned by the n -truncated anima, and $Cat_n \subset Cat_{\infty}$ for the full subcategory spanned by the n -categories.

Ex $An_{\leq n}$ and Cat_n are $(n+1)$ -categories.

Def Let $f: I \rightarrow J$ be a functor of ∞ -categories and $n \in \mathbb{N} \cup \{\infty\}$. We say that f is an n -lim-equivalence if for every n -category $\mathcal{C}$ and functor $g: J \rightarrow \mathcal{C}$, the following hold:

- (1) The limit $\lim_J g$ exists if and only if $\lim_I gf$ exists.
- (2) If both of these limits exist, then the natural map

$$\lim_J g \longrightarrow \lim_I gf$$

is an equivalence.

> If $n = \infty$, we just call this a lim-equivalence.

Prop. The inclusion $\Delta^{<n} \subset \Delta$ is a lim-equivalence.

Prop. For each $n \geq 0$, the inclusions $\Delta_{\leq n}^{<n} \subset \Delta^{<n}$ and $\Delta_{\leq n} \subset \Delta$ are n -lim-equivalences, but not $(n+1)$ -lim-equivalences.

Upshot. We can always formulate the sheaf condition using a whole cosimplicial diagram.

- > For sheaves valued in a 1-category (like Set), this is equivalent to being the limit over the first two terms
- > For sheaves valued in a 2-category (like Cat_1), this is equivalent to being the limit over the first three terms
 - The 3rd term encodes the cocycle condition

Actions & quotients

Def Let $\mathcal{C}$ be an ∞ -category with finite products and let $M: \Delta^{\text{op}} \rightarrow \mathcal{C}$ be a monoid object. A left M -module (or object of $\mathcal{C}$ with a left M -action) is a map of simplicial objects $p: A \rightarrow M$ such that for each $n \geq 0$, the maps

$$p([n]): A([n]) \rightarrow M([n]) \quad \text{and} \quad A([n]) \rightarrow A(\{n\})$$

$\downarrow$
 A_n

$\downarrow$
 $M_n \simeq M_1^{\times n}$

$\downarrow$
 A_n

$\downarrow$
 A_0

exhibit $A([n])$ as the product $M([n]) \times A(\{n\})$.

- > In this case, we say that M acts on the 0-simplices A_0 .
- > Given $X \in \mathcal{C}$ a left action of M on $\mathcal{C}$ consists of a left M -module A together with an equivalence $A_0 \simeq X$.
- > We write $\text{LMod}_M(\mathcal{C}) \subset \text{Fun}(\Delta^{\text{op}}, \mathcal{C})_M$ for the full subcat. spanned by the left M -modules.

Note. The axioms guarantee that A is of the form

$$\cdots \begin{array}{c} \xrightarrow{\quad} \\ \xrightarrow{\quad} \\ \xrightarrow{\quad} \\ \xrightarrow{\quad} \end{array} M_1 \times M_1 \times A_0 \begin{array}{c} \xrightarrow{\quad} \\ \xrightarrow{\quad} \\ \xrightarrow{\quad} \\ \xrightarrow{\quad} \end{array} M_1 \times A_0 \begin{array}{c} \xrightarrow{a} \\ \xrightarrow{\quad} \\ \xrightarrow{\quad} \\ \xrightarrow{\text{pr}_2} \end{array} A_0.$$

the action map

Def. Let $M: \Delta^{\text{op}} \rightarrow \mathcal{C}$ and let $A: \Delta^{\text{op}} \rightarrow \mathcal{C}$ be an action of M on an object $X \in \mathcal{C}$. The quotient X/M is the colimit (provided it exists)

$$X/M = \operatorname{Colim}_{\Delta^{\text{op}}} A$$

Note. In the special case $\mathcal{C} = \text{Set}$, we see that we're looking at the coequalizer

$$\operatorname{Coeq} \left(M \times X \begin{array}{c} \xrightarrow{\text{act}} \\ \xrightarrow{\text{pr}_2} \end{array} X \right)$$

which is the quotient of X by the equivalence relation generated by $m \cdot x \sim x$ for all $m \in M$ and $x \in X$. So indeed this definition of X/M recovers the usual definition of the quotient of X by M .

Coming Soon Given a scheme X with an action of a monoid scheme, the quotient of X by M in $\text{Fun}_{\text{fpqc}}(\text{CAlg}, \text{Set})$ is not always interesting/well-behaved (e.g., if $X = *$). Instead, we'll consider the quotient stack X/M in $\text{Fun}_{\text{fpqc}}(\text{CAlg}, \text{An})$.

- > Even when $X = *$, this is interesting.
- > We'll use this to define equivariant sheaves.